BLACK PEN Answer all questions first. **GREEN PEN** Then: tick and rewrite mistakes.

LESSON 058 · Iteration: WHILE Loops

Name: _____ Class: _____ Date: _____

Work through the pages in order. Answer in BLACK PEN first. Check in GREEN PEN only after the Exit Ticket.

Do Now · Recall previous learning

Q1. When is a count-controlled loop suitable?

Q2. Why use pseudocode?
A. Run directly on the CPU
B. Plan steps without exact syntax
C. Replace all program testing
D. Draw the user interface

Q3. “A diamond is used only for unconditional calculations.” Correct this. Explain why.

Q4. A program works out a bill, then applies a discount. “The order of steps never matters.” Correct this. What could go wrong?

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I Do · Read new knowledge

Key vocabulary

WHILE loop — Checks a condition before each iteration; repeats while it is true.

Condition — A comparison such as attempts < 3 controls continuation.

Update — Change loop variables so the condition can eventually become false.

Unknown repetition — WHILE suits a task where the number of tries is not fixed beforehand.

Algorithm — A sequence of precise problem-solving instructions.

Trace table — A table tracking variable values during execution.

New learning

WHILE loop: Checks a condition before each iteration; repeats while it is true. For example, consider a password attempt counter.

Condition: A comparison such as attempts < 3 controls continuation. For example, consider a quiz retry.

Update: Change loop variables so the condition can eventually become false. For example, consider a low stock checker.

Unknown repetition: WHILE suits a task where the number of tries is not fixed beforehand. For example, consider a temperature alarm.

Worked example

A password attempt counter is being developed. Explain how while loop and condition are used. [4 marks]

1. Identify where this particular program needs control or values.

2. WHILE loop [1]; Checks a condition before each iteration; repeats while it is true. [1].

3. Condition [1]; A comparison such as attempts < 3 controls continuation. [1].

4. Give one accurate point and a linked reason for each technique.

We Do · Practise with guidance

Guided 1

A quiz retry is being programmed. Explain why while loop matters. [2 marks]

Worked method: WHILE loop [1]; Checks a condition before each iteration; repeats while it is true. [1].

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Guided 2

For a low stock checker, explain how condition is used. [2 marks]

Worked method: Condition [1]; A comparison such as attempts < 3 controls continuation. [1].

Guided 3

A temperature alarm needs update and unknown repetition. Explain the roles of both. [4 marks]

Worked method: Update [1]; Change loop variables so the condition can eventually become false. [1]. Unknown repetition [1]; WHILE suits a task where the number of tries is not fixed beforehand. [1].

BLACK PEN Answer all questions first. **GREEN PEN** Then: tick and rewrite mistakes.

You Do · Work independently

BLACK PEN: Write your own answers. Do not look at the Review & Improve pages yet.

Code First · Predict → Write → Test → Improve

Add one line so the loop reads another answer and can finish.

```
answer = input("Continue? Y/N")
while answer != "N"
    ...
endwhile
print("Stopped")
```

write

Question 1

State TWO concepts needed for this programming technique. [2 marks] Source: Original J277-style teaching question; not an OCR past-paper original.

Question 2

Explain how condition and update can help a correction prompt. [4 marks] Source: Original J277-style teaching question; not an OCR past-paper original.

Question 3

Explain the roles of while loop, update and unknown repetition when creating a survey response. [6 marks] Source: Original J277-style teaching question; not an OCR past-paper original.

BLACK PEN Answer all questions first. **GREEN PEN** Then: tick and rewrite mistakes.

Exit Ticket · Answer before checking

Checks a condition before each iteration; repeats while it is true.

BLACK PEN Answer all questions first. **GREEN PEN** Then: tick and rewrite mistakes.

Review & Improve · GREEN PEN

Now change to GREEN PEN. Find each original question. Tick correct answers; read the model and rewrite mistakes.

Page 1 onwards · Do Now

Q1. When the number of repetitions is known or determined by a fixed count.

Q2. B — Plan steps without exact syntax

Q3. A decision or condition with alternative paths.

Q4. Steps must be clear and in the right order. Check that conditions give the right result for every allowed input.

I Do / We Do pages · Guided practice

Q1. WHILE loop [1]; Checks a condition before each iteration; repeats while it is true. [1].

Q2. Condition [1]; A comparison such as attempts < 3 controls continuation. [1].

Q3. Update [1]; Change loop variables so the condition can eventually become false. [1]. Unknown repetition [1]; WHILE suits a task where the number of tries is not fixed beforehand. [1].

You Do pages · Code First

```
answer = input("Continue? Y/N")
while answer != "N"
    answer = input("Continue? Y/N")
endwhile
print("Stopped")
```

Expected result: Stopped (test input sequence: Y, then N)

You Do pages · Independent questions

Q1. WHILE loop [1]; Update [1].

Q2. Condition [1]; A comparison such as attempts < 3 controls continuation. [1]. Update [1]; Change loop variables so the condition can eventually become false. [1].

Q3. WHILE loop [1]; Checks a condition before each iteration; repeats while it is true. [1]. Update [1]; Change loop variables so the condition can eventually become false. [1]. Unknown repetition [1]; WHILE suits a task where the number of tries is not

fixed beforehand. [1].

Exit Ticket · Model answer

True

Try the question again

Choose one question you got wrong. All correct? Choose the hardest.

1. Copy the whole original question:

2. Write a complete new answer in GREEN PEN. Show working if needed: